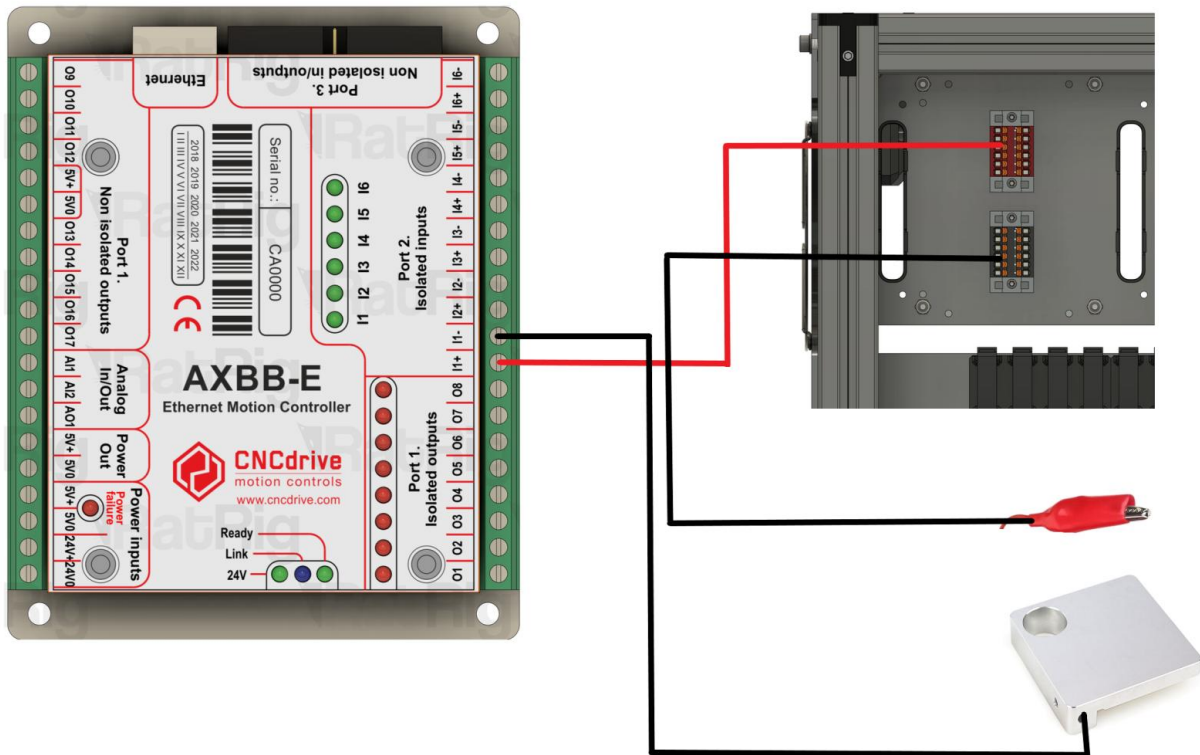


AXBB-E_Probe_Wiring_Guide

Option 1: Two-Wire Setup (Isolated Probe)

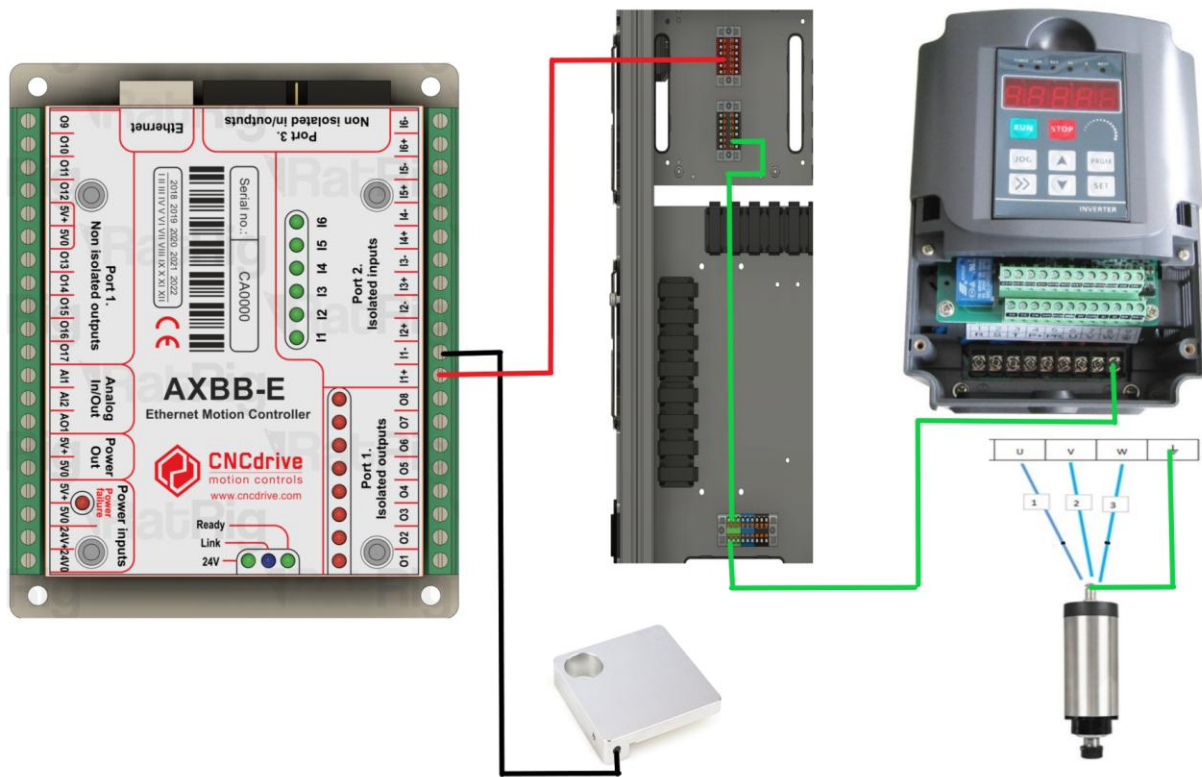


Select an unused input on the AXBB motion controller (e.g., input I1).

- Connect the **I1+** terminal directly to the **24V+** power rail (red wire).
- Connect the **alligator clip** directly to the **0V** rail (black wire).
- Connect the **probe plate** to the **I1-** terminal (black wire).

Note: This wiring method requires a two-wire cable extending from the control cabinet to the probe plate and alligator clip. The clip **must** be manually attached to the cutting tool prior to initiating any probing cycle.

Option 2: Single-Wire Setup (Grounded Spindle)



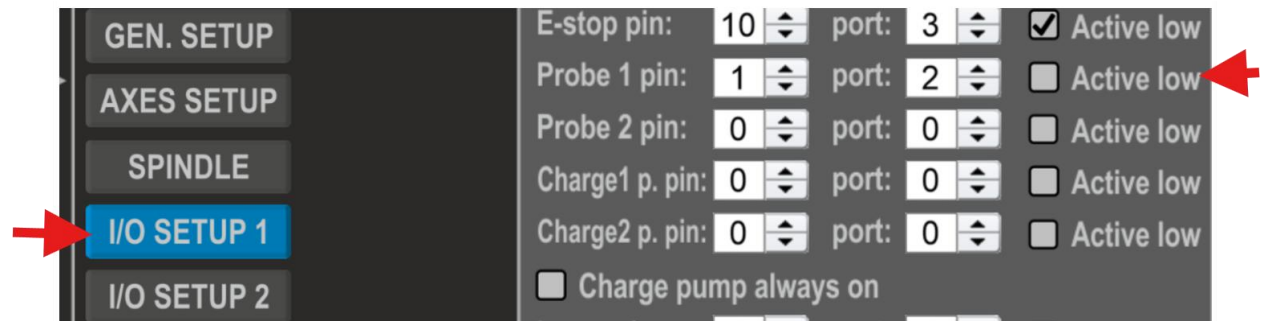
We will again use input I1 in this example.

- Connect the **I1+** terminal directly to the **24V+** power rail (red wire).
- Connect the **probe plate** to the **I1-** terminal (black wire).
- Connect the **0V** rail directly to the **ground (PE)** rail (green wire).
- Connect the **ground rail** to the **ground terminal** on the VFD / inverter (green wire).

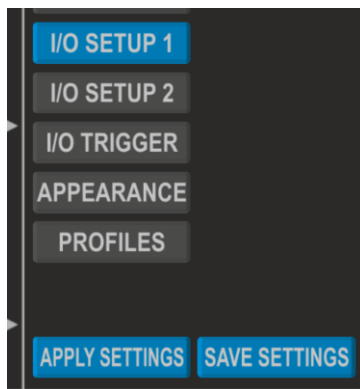
The ground terminal on the inverter must be securely connected to the shielding of the spindle power cable. Inside the spindle connector, this shielding connects directly to the spindle's metallic body.

With this common ground configuration, all metallic machine components are grounded and connected to 0V. Since 0V is continuously present at the spindle and tool, only a single wire is needed between the control cabinet and the probe plate.

Setting Up the Probe Input in UCCNC:

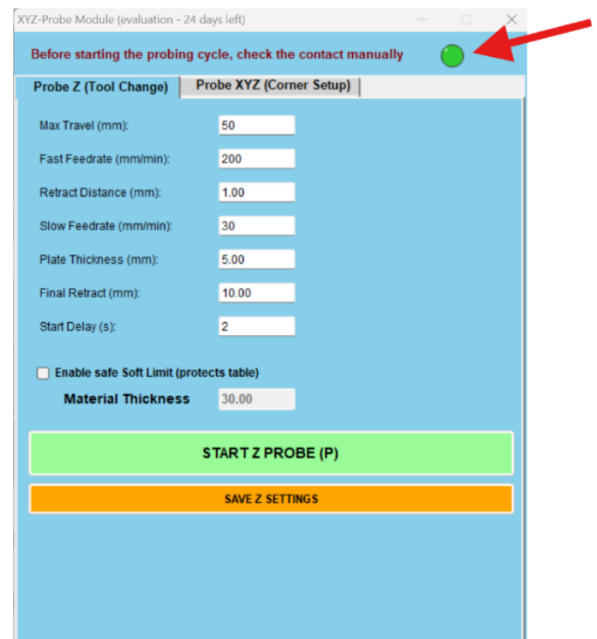


- In the **I/O SETUP 1** tab, locate **Probe 1 pin**.
- Set the pin to **1** and the port to **2** (corresponding to input I1 on Port 2).
- Ensure the **Active low** checkbox matches your wiring logic (as shown in the reference image).
- **Important:** Press **Enter** on your keyboard after entering each numerical value, as UCCNC will not register changes otherwise.
- Click **APPLY SETTINGS** followed by **SAVE SETTINGS** to permanently store the configuration.



CRITICAL PRE-RUN SAFETY CHECK: Before running any automatic probing cycle with the ProbeXYZ software, **always test contact manually**:

- Touch the touch plate directly against the CNC tool/spindle.
- Verify that the green indicator LED lights up in the software and an audible beep sounds.
- *Tip:* Keep your hand near the E-Stop button during the first test cycle in case the axis fails to halt upon contact.



If you encounter any issues or have questions regarding the ProbeXYZ software, feel free to contact our technical support team via emails:

- Support Email 1: drago.grgic3@gmail.com
- Support Email 2: drago.grgic2@st.t-com.hr